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University of Utah Spring 2026

MATH 2270-006
Midterm 3 Questions

Instructor: Alp Uzman

April 10 2026, 9:40 AM - 10:30 AM

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1. **[points]** Consider the following matrix:

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}.$$

(a) **[20 points]** Compute the scalars a_0, a_1 and a_2 such that the characteristic polynomial of A is $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$.

$$\text{char}_A(\lambda) = \det(\lambda I - A) = \det\begin{pmatrix} \lambda-2 & -3 & -1 \\ -4 & \lambda & 1 \\ 0 & -2 & \lambda-1 \end{pmatrix}$$

$$= (\lambda-2)\lambda(\lambda-1) + 0\cdot(-3)\cdot 1 + (-4)(-2)(-1)$$

$$- (-1)\cdot\lambda\cdot 0 - (-3)(-4)(\lambda-1) - (\lambda-2)(-2)\ 1$$

$$= \lambda(\lambda-1)(\lambda-2) - 8 - 12(\lambda-1) + 2(\lambda-2)$$

$$= (\lambda^2-\lambda)(\lambda-2) - 8 - 12\lambda + 12 + 2\lambda - 4$$

$$= \lambda^3 - 2\lambda^2 - \lambda^2 + 2\lambda - 8 - 12\lambda + 12 + 2\lambda - 4$$

$$= \lambda^3(1) + \lambda^2(-2-1) + \lambda(2-12+2) + 1(-8+12-4)$$

$$= \boxed{\lambda^3 + (-3)\lambda^2 + (-8)\lambda + (0)\,1}$$

$$\boxed{\begin{aligned} a_2 &= -3 \\ a_1 &= -8 \\ a_0 &= 0 \end{aligned}}$$

(b) **[20 points]** Compute all eigenvalues of A.

$9+32$
$=41$

$$\mathrm{char}_A(\lambda) = \lambda^3 - 3\lambda^2 - 8\lambda$$

$$= \lambda(\lambda^2 - 3\lambda - 8)$$

$$= \lambda\left(\lambda^2 + 2\cdot\left(\frac{-3}{2}\right)\lambda + \left(\frac{-3}{2}\right)^2 - \left(\frac{-3}{2}\right)^2 - 8\right)$$

$$= \lambda\left[\left(\lambda - \frac{3}{2}\right)^2 - \left(\frac{\sqrt{41}}{2}\right)^2\right]$$

$$= \lambda\left(\lambda - \frac{3}{2} - \frac{\sqrt{41}}{2}\right)\left(\lambda - \frac{3}{2} + \frac{\sqrt{41}}{2}\right)$$

$$= \lambda\left(\lambda - \frac{3+\sqrt{41}}{2}\right)\left(\lambda - \frac{3-\sqrt{41}}{2}\right).$$

$$\Rightarrow \boxed{\mathrm{Spec}(A) = \left\{0, \frac{3+\sqrt{41}}{2}, \frac{3-\sqrt{41}}{2}\right\}}.$$

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(c) **[5 points]** Compute two 3×3 matrices P and D such that all of the following are satisfied simultaneously:

- $A = PDP^{-1}$,
- P is invertible, and
- D is diagonal.

$\lambda_1 = 0,\ \lambda_2 = \frac{3+\sqrt{41}}{2},\ \lambda_3 = \frac{3-\sqrt{41}}{2}.$

$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}.\quad P = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix}.$

$\boxed{\lambda_1 = 0}$ $\begin{pmatrix} 2 & 3 & 1 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix} \to \begin{pmatrix} 6 & 3 & 0 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix} \to \begin{pmatrix} 2 & 1 & 0 \\ 4 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix} \to \begin{pmatrix} 2 & 1 & 0 \\ 0 & -2 & -1 \\ 0 & 2 & 1 \end{pmatrix}$

$\downarrow$

$\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$\boxed{\begin{aligned} 2x + y &= 0 \\ 2y + z &= 0 \\ z &\text{ free.} \end{aligned}} \Rightarrow \boxed{v_1 = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}}$

$\boxed{\lambda_2 = \frac{3+\sqrt{41}}{2}}$

$AM = GM = 1$

$\begin{pmatrix} \frac{3+\sqrt{41}}{2} - 2 & 3 & 1 \\ 4 & \frac{3+\sqrt{41}}{2} & 1 \\ 0 & -2 & \frac{3+\sqrt{41}}{2} - 1 \end{pmatrix} = \begin{pmatrix} \frac{-1+\sqrt{41}}{2} & -3 & -1 \\ -4 & \frac{3+\sqrt{41}}{2} & 1 \\ 0 & -2 & \frac{1+\sqrt{41}}{2} \end{pmatrix}$

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$(3+\sqrt{41})(1+\sqrt{41}) = 3+4\sqrt{41}+41 = 44+4\sqrt{41}$

$(-3+\sqrt{41})(1+\sqrt{41}) = -3-2\sqrt{41}+41 = 38-2\sqrt{41}$

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$$\to \left(\begin{array}{c|c|c} -1+\sqrt{41} & -6 & -2 \\ \hline -8 & 3+\sqrt{41} & 2 \\ \hline 0 & -4 & 1+\sqrt{41} \end{array}\right) \to \left(\begin{array}{c|c|c} -9+\sqrt{41} & -3+\sqrt{41} & 0 \\ \hline -8 & 3+\sqrt{41} & 2 \\ \hline 0 & -4 & 1+\sqrt{41} \end{array}\right)$$

$$\left(\begin{array}{c|c|c} -9+\sqrt{41} & -3+\sqrt{41} & 0 \\ \hline -8-8\sqrt{41} & 52+4\sqrt{41} & 0 \\ \hline 0 & -8 & 2+2\sqrt{41} \end{array}\right) \leftarrow \left(\begin{array}{c|c|c} -9+\sqrt{41} & -3+\sqrt{41} & 0 \\ \hline -8-8\sqrt{41} & 44+4\sqrt{41} & 2+2\sqrt{41} \\ \hline 0 & -8 & 2+2\sqrt{41} \end{array}\right)$$

$$\Rightarrow \boxed{\begin{array}{l} (-9+\sqrt{41})x + (-3+\sqrt{41})y = 0 \\ -\cancel{8}^{4}y + (\cancel{2}^{1}+\cancel{2}\sqrt{41})z = 0 \\ z \text{ free.} \end{array}} \Rightarrow \begin{pmatrix} -\dfrac{(-3+\sqrt{41})(1+\sqrt{41})}{-9+\sqrt{41}} \\ 1+\sqrt{41} \\ 4 \end{pmatrix}$$

$$\Rightarrow \boxed{= \begin{pmatrix} \dfrac{-38+2\sqrt{41}}{-9+\sqrt{41}} \\ 1+\sqrt{41} \\ 4 \end{pmatrix}} = \begin{pmatrix} \dfrac{13+\sqrt{41}}{2} \\ 1+\sqrt{41} \\ 4 \end{pmatrix} \Rightarrow \boxed{v_2 = \begin{pmatrix} 13+\sqrt{41} \\ 2+2\sqrt{41} \\ 8 \end{pmatrix}}$$

$\lambda_3 = \dfrac{3-\sqrt{41}}{2}$ AM = GM = 1

$$\boxed{v_3 = \begin{pmatrix} 13-\sqrt{41} \\ 2-2\sqrt{41} \\ 8 \end{pmatrix}}$$

$$\begin{array}{r} 7 \\ 38 \\ \times\ 9 \\ \hline 342 \\ 342 \\ \hline 260 \end{array}$$

$(-9+\sqrt{41})(-9-\sqrt{41}) = 81-41 = 40.$

$(-38+2\sqrt{41})(-9-\sqrt{41}) = 38\cdot 9 + 20\sqrt{41} - 82 = 260+20\sqrt{41}$

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2. **[20 points]** Consider the following matrix:

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix}.$$

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(a) **[5 points]** Compute all eigenvalues of A.

$\boxed{\text{Spec}(A) = \{2, -1, 3\}.}$

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(b) **[5 points]** For each distinct eigenvalue of A compute a basis of the associated eigenspace.

$\boxed{\lambda_1 = 2}$ $\Rightarrow$ basis for $E(2)$:

$$\left(e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right).$$

$\boxed{\lambda_2 = -1}$ $\Rightarrow$ basis for $E(-1)$:

$$\left(e_5 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right).$$

$\lambda_3 = 3.$ $B = \begin{pmatrix} -1 & 1 \\ 0 & 3 \end{pmatrix}.$ $\begin{pmatrix} 3+1 & -1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 0 & 0 \end{pmatrix}.$

$$\Rightarrow \left(v_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 4 \end{pmatrix} \right)$$ basis for $E(3)$!

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(c) **[5 points]** Compute the algebraic multiplicities of each distinct eigenvalue of A.

λ :	2	-1	3
AM	4	+1	1

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(d) **[5 points]** Compute the geometric multiplicities of each distinct eigenvalue of A.

λ	2	-1	3
GM	2	1	1

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3. **[31 points]** Consider the following three vectors in $\mathbb{R}^3$:

$$x = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \quad y = \begin{pmatrix} 4 \\ -8 \\ 0 \end{pmatrix}, \quad z = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

and let W be the span of x and y. Perform the following tasks:

(a) **[20 points]** Compute the 3×3 matrix that corresponds to orthogonal projection onto the span of x.

$\langle x, x \rangle = 9 \Rightarrow |x| = 3.$

$$\text{proj}_x = \left(\frac{x}{|x|}\right)\left(\frac{x}{|x|}\right)^T = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix} \begin{pmatrix} 2/3 & 1/3 & 2/3 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} 4/9 & 2/9 & 4/9 \\ 2/9 & 1/9 & 2/9 \\ 4/9 & 2/9 & 4/9 \end{pmatrix}}$$

$y = \begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix}$

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(6) **[5 points]** Compute the orthogonal projection of z onto W.

$\langle y, z \rangle = 16 + 64 + 1 = 81 \Rightarrow |y| = 9.$

$\mathrm{proj}_W z = \frac{y}{|y|} \left(\frac{y}{|y|}\right)^T = \begin{pmatrix} 4/9 \\ 8/9 \\ 1/9 \end{pmatrix} \begin{pmatrix} 4/9 & 8/9 & 1/9 \end{pmatrix}$

$= \begin{pmatrix} 16/81 & 32/81 & 4/81 \\ 32/81 & 64/81 & 8/81 \\ 4/81 & 8/81 & 1/81 \end{pmatrix}$

$\mathrm{proj}_W z = \mathrm{proj}_W x + \mathrm{proj}_W y = \begin{pmatrix} 1/9 & 2/9 & 4/9 \\ 2/9 & 4/9 & 8/9 \\ 4/9 & 8/9 & 1/9 \end{pmatrix} + \begin{pmatrix} 16/81 & 32/81 & 4/81 \\ 32/81 & 64/81 & 8/81 \\ 4/81 & 8/81 & 1/81 \end{pmatrix}$

$= \begin{pmatrix} \frac{34}{81} & \frac{18+32}{81} & \frac{36+4}{81} \\ \frac{18+32}{81} & \frac{32+64}{81} & \frac{72+8}{81} \\ \frac{36+4}{81} & \frac{72+8}{81} & \frac{4+1}{81} \end{pmatrix} = \begin{pmatrix} \frac{34}{81} & \frac{50}{81} & \frac{40}{81} \\ \frac{50}{81} & \frac{96}{81} & \frac{80}{81} \\ \frac{40}{81} & \frac{80}{81} & \frac{5}{81} \end{pmatrix}$

$x = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ $y = \begin{pmatrix} 4 \\ -8 \\ 0 \end{pmatrix}$

(b) **[6 points]** Compute an orthonormal basis of W.

$\langle x, x \rangle = 9$

$\langle x, y \rangle = 0 \Rightarrow (x, y)$ OB for W.

$\langle y, y \rangle = 80 = 4^2 \cdot 5$

$\Rightarrow \left(\frac{x}{|x|}, \frac{y}{|y|} \right)$ ONB for W.

$\frac{x}{|x|} = \begin{pmatrix} 2/3 \\ 1/3 \\ 2/3 \end{pmatrix}$. $\frac{y}{|y|} = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \\ 0 \end{pmatrix}$.

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$80 = 5 \cdot 20 = 5 \cdot 5 \cdot 5$

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$\boxed{3.6}$ $x = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$, $y = \begin{pmatrix} 5 \\ -8 \\ 0 \end{pmatrix}$, $z = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

$\langle x, x \rangle = 9$ $\quad |x| = 3$

$\langle y, y \rangle = 16 + 64 = 80$ $\quad |y| = \sqrt{80} = 4\sqrt{5}$

$\langle x, y \rangle = 0. \Rightarrow \boxed{x \perp y}$

$\langle x, z \rangle = -4 + 1 + 2 = -1$

$\langle y, z \rangle = -8 - 8 = -16.$

$\text{proj}_W(z) = \frac{\langle z, x \rangle}{\langle x, x \rangle} x + \frac{\langle z, y \rangle}{\langle y, y \rangle} y$

$= \frac{-1}{9} x + \frac{-16}{80} y$

$= \frac{-1}{9} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \frac{-1}{5} \begin{pmatrix} 5 \\ -8 \\ 0 \end{pmatrix}$

$= \frac{-5}{45} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \frac{-9}{45} \begin{pmatrix} 5 \\ -8 \\ 0 \end{pmatrix}$

$= \frac{-1}{45} \begin{pmatrix} 10 + 36 \\ 5 - 72 \\ 10 + 0 \end{pmatrix}$

$= \boxed{\frac{1}{45} \begin{pmatrix} -46 \\ 67 \\ -10 \end{pmatrix}}$

4. **[4 points]** In this problem, let A be an arbitrary square matrix.

(a) **[2 points]** The i-th row sum of A is by definition the sum of all the entries in the i-th row of A. Verify that if each row sum of A is equal to the same number r, then r is an eigenvalue of A.

$$\mathbb{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \Rightarrow A\mathbb{1} = \begin{pmatrix} \text{1st row sum} \\ \text{2nd row sum} \\ \vdots \\ \text{nth row sum} \end{pmatrix}$$

$$= \begin{pmatrix} r \\ r \\ \vdots \\ r \end{pmatrix} = r\mathbb{1}. \checkmark$$

(b) **[2 points]** The j-th column sum of A is by definition the sum of all the entries in the j-th column of A. Verify that if each column sum of A is equal to the number c, then c is an eigenvalue of A.

$$A^T\mathbb{1} = \begin{pmatrix} \text{1st col. sum} \\ \text{2nd col. sum} \\ \vdots \\ \text{nth col. sum} \end{pmatrix} = \begin{pmatrix} c \\ c \\ \vdots \\ c \end{pmatrix}$$

$= c\mathbb{1} \Rightarrow c$ eigenvalue of A^T.

$\text{char}_{A^T}(\lambda) = \det(\lambda I - A^T) = \det((\lambda I - A)^T)$

$= \det(\lambda I - A) = \text{char}_A(\lambda) \Rightarrow c$ eigenvalue of A. $\checkmark$

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